

Initiating Search

November 5, 2025, 8:49 PM

• Search:

CAS Lexicon Search:

Thioureas - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (4,222)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (1,804)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	
Language:	English	
Publication Year:	2020 to 2025	

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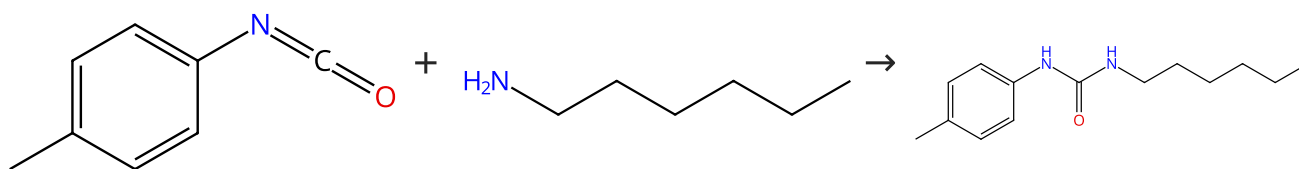
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (71)

 Suppliers (71)

 Suppliers (2)

31-316-CAS-20805913

Steps: 1 Yield: 100%

1.1 2 - 3 min, 0 °C

Experimental Protocols

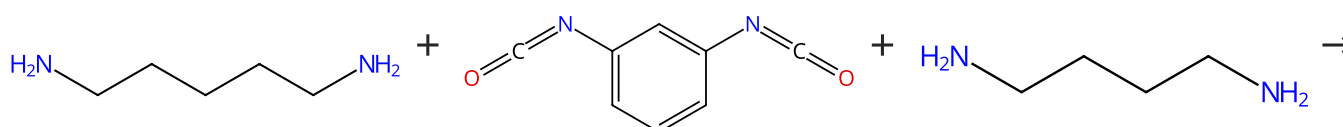
Facile synthesis of phthalidyl fused spiro thiohyd antioins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas

By: Mandal, Subhro; et al

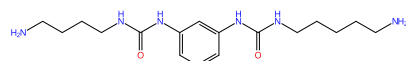
Tetrahedron (2020), 76(2), 130817.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (71)

 Suppliers (57)

 Suppliers (67)


31-614-CAS-37221630

Steps: 1 Yield: 100%

1.1 **Reagents:** Picric acid
Catalysts: 2,2',2''-[[7,15,23-Tris(1,1-dimethylethyl)-3,11,19-trioxatetracyclo[19.3.1.1^{5,9}.1^{13,17}]heptacosa-1(25),5,7,9(27),13,15,17(26),21,23-nonaene-25,26,27-triyl]tris(oxy)]tris[*N,N*-diethylacetamide]
Solvents: Chloroform-*d*, Acetonitrile-*d*₃

1.2 5 min, rt

1.3 **Reagents:** Picric acid
Catalysts: 2,2',2''-[[7,15,23-Tris(1,1-dimethylethyl)-3,11,19-trioxatetracyclo[19.3.1.1^{5,9}.1^{13,17}]heptacosa-1(25),5,7,9(27),13,15,17(26),21,23-nonaene-25,26,27-triyl]tris(oxy)]tris[*N,N*-diethylacetamide]
Solvents: Chloroform-*d*, Acetonitrile-*d*₃

1.4 **Reagents:** 1,8-Diazabicyclo[5.4.0]undec-7-ene

Experimental Protocols

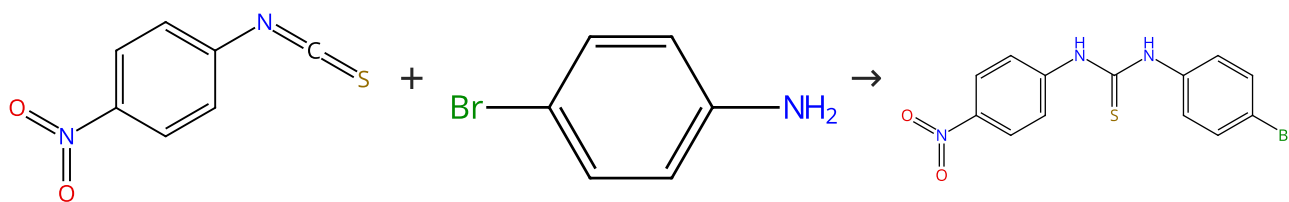
Supramolecular protection with a recyclable molecular container: an efficient strategy for the one-pot selective functionalization of polyfunctional substrates

By: Lambert, Simon; et al

Organic Chemistry Frontiers (2023), 10(17), 4230-4242.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

Suppliers (83)

Supplier (1)

31-614-CAS-36765081

Steps: 1 Yield: 100%

Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders

By: Minneci, Marco; et al

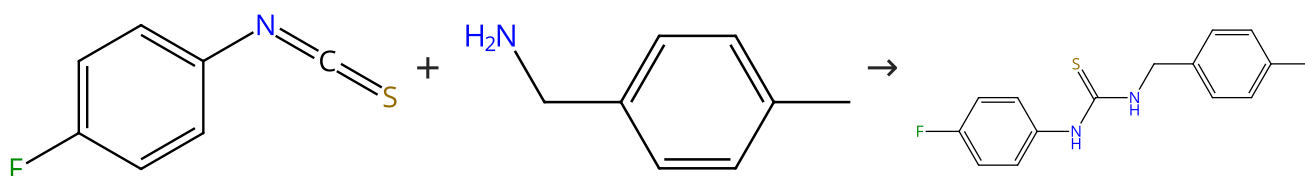
Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

1.1 Solvents: (1*S*,5*R*)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt

Experimental Protocols

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (57)

Suppliers (80)

Supplier (1)

31-316-CAS-20805857

Steps: 1 Yield: 100%

Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas

By: Mandal, Subhro; et al

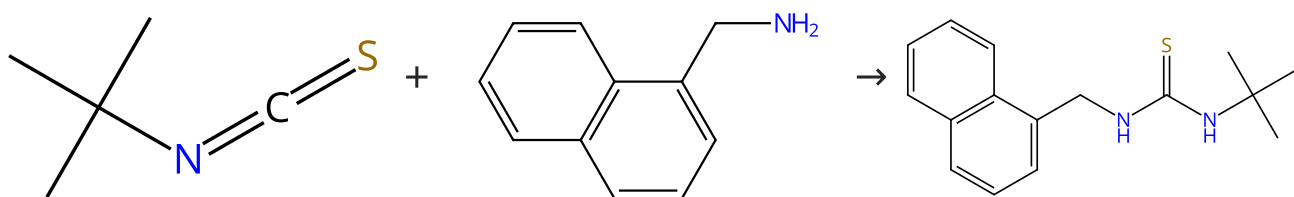
Tetrahedron (2020), 76(2), 130817.

1.1 2 - 3 min, 0 °C

Experimental Protocols

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (42)

Suppliers (89)

31-614-CAS-24399556

Steps: 1 Yield: 100%

Gram Scalable Method to Synthesize Biscarbodiimides and Asymmetric Monocarbodiimides: A Platform to Access an Array of Phosphaguanidines, Amidines, and Guanidines

By: Camelio, Andrew M.; et al

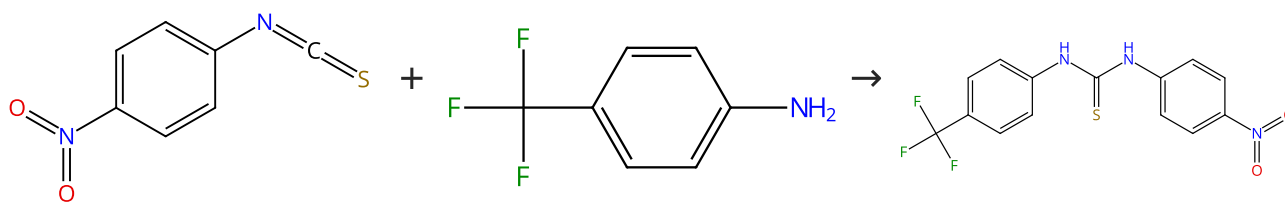
Journal of Organic Chemistry (2022), 87(4), 2022-2044.

1.1 Solvents: Diethyl ether; 0 °C; 24 h, 0 °C → 23 °C

Experimental Protocols

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

Suppliers (86)

Supplier (1)

31-614-CAS-36765085

Steps: 1 Yield: 100%

1.1 Solvents: (1*S*,5*R*)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt
Experimental Protocols

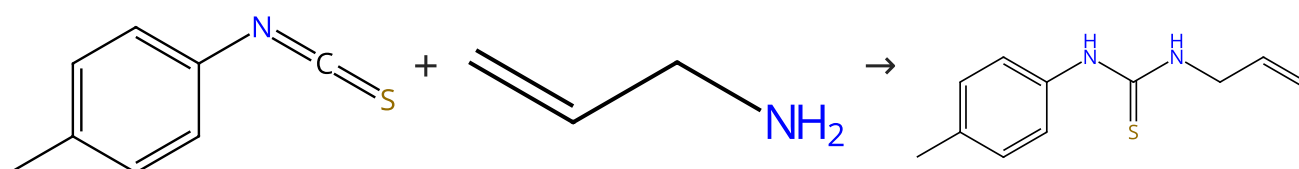
Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders

By: Minneci, Marco; et al

Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

Scheme 7 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

Suppliers (33)

Suppliers (14)

31-316-CAS-20805852

Steps: 1 Yield: 100%

1.1 2 - 3 min, 0 °C
Experimental Protocols

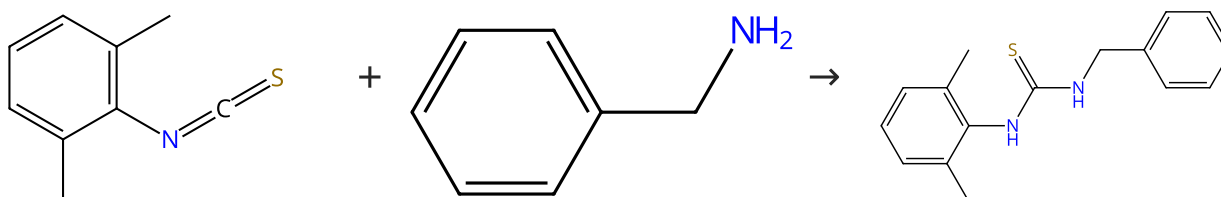
Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas

By: Mandal, Subhro; et al

Tetrahedron (2020), 76(2), 130817.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (72)

Suppliers (81)

Suppliers (5)

31-614-CAS-24399545

Steps: 1 Yield: 100%

1.1 Solvents: Diethyl ether; 0 °C; 12 h, 0 °C → 23 °C
Experimental Protocols

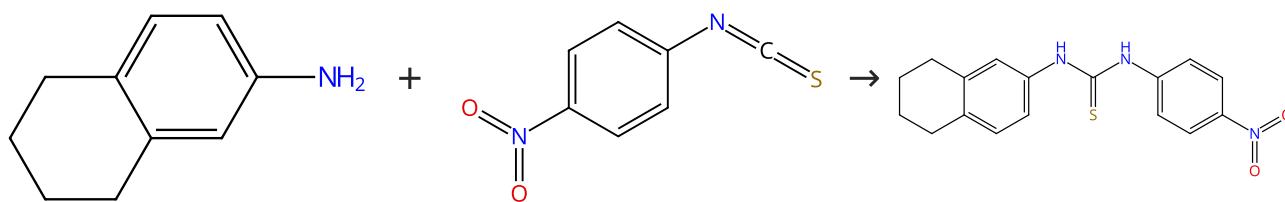
Gram Scalable Method to Synthesize Biscarbodiimides and Asymmetric Monocarbodiimides: A Platform to Access an Array of Phosphaguanidines, Amidines, and Guanidines

By: Camelio, Andrew M.; et al

Journal of Organic Chemistry (2022), 87(4), 2022-2044.

Scheme 9 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (61)

Suppliers (75)

31-614-CAS-36765091

Steps: 1 Yield: 100%

Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders

By: Minneci, Marco; et al

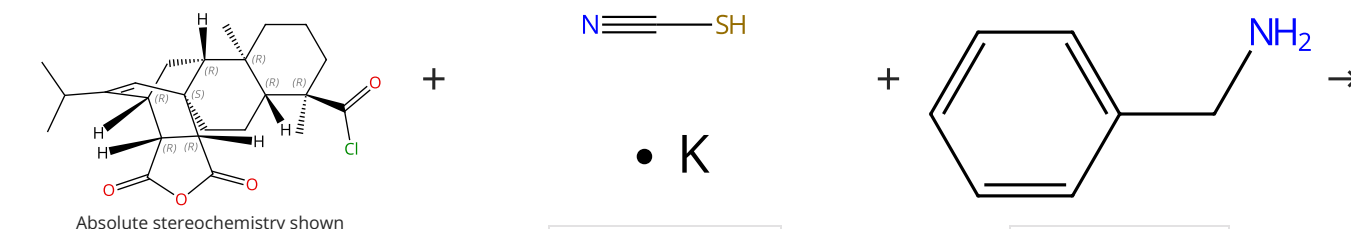
Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

1.1 Solvents: (1*S*,5*R*)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt

Experimental Protocols

Scheme 10 (1 Reaction)

Steps: 1 Yield: 100%

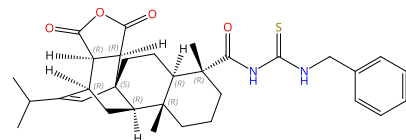


Absolute stereochemistry shown

Supplier (1)

Suppliers (116)

Suppliers (81)

Absolute stereochemistry shown,
Rotation (+)

31-614-CAS-34878858

Steps: 1 Yield: 100%

Synthesis of New Substituted Ureides and Thioureides Based on Maleopimaric Acid

By: Vafina, G. F.; et al

Chemistry of Natural Compounds (2022), 58(6), 1055-1061.

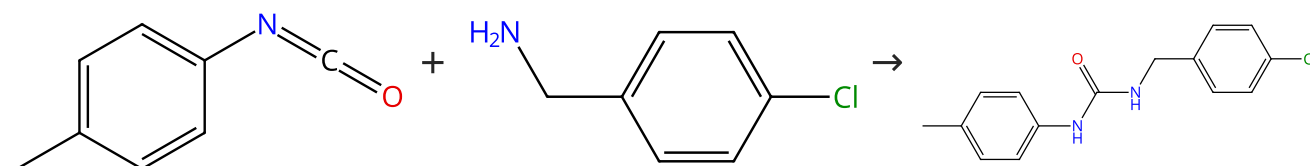
1.1 Solvents: Acetone; rt; 40 min, reflux; reflux → 5 °C

1.2 Solvents: Acetone; 0 - 5 °C; rt

Experimental Protocols

Scheme 11 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (71)

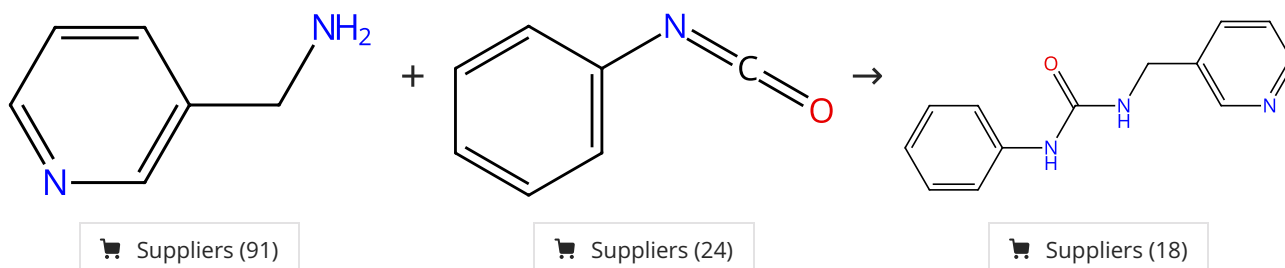
Suppliers (87)

Suppliers (8)

31-316-CAS-20805911 Steps: 1 Yield: 100% 1.1 2 - 3 min, 0 °C Experimental Protocols	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
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Scheme 12 (1 Reaction)

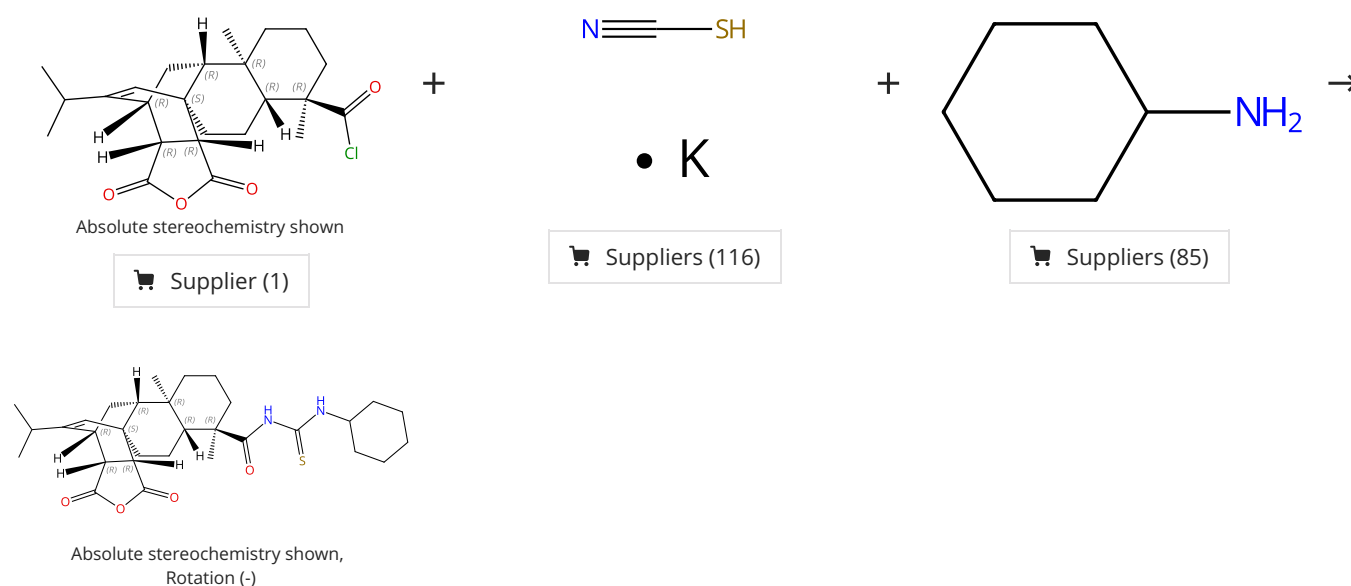
Steps: 1 Yield: 100%



31-316-CAS-20805909 Steps: 1 Yield: 100% 1.1 2 - 3 min, 0 °C Experimental Protocols	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
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Scheme 13 (1 Reaction)

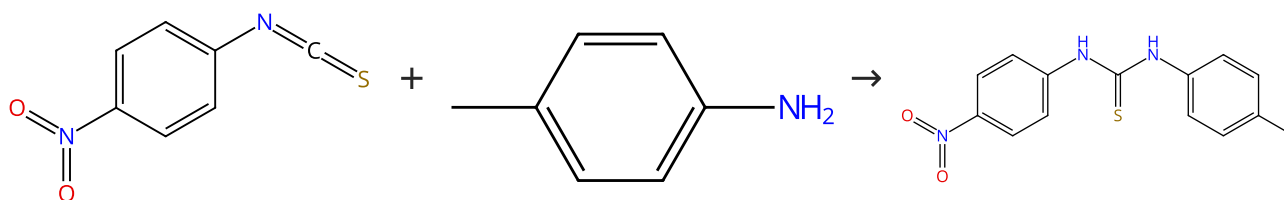
Steps: 1 Yield: 100%



31-614-CAS-34878853 Steps: 1 Yield: 100% 1.1 Solvents: Acetone; rt; 40 min, reflux; reflux → 5 °C 1.2 Solvents: Acetone; 0 - 5 °C; rt Experimental Protocols	Synthesis of New Substituted Ureides and Thioureides Based on Maleopimaric Acid By: Vafina, G. F.; et al Chemistry of Natural Compounds (2022), 58(6), 1055-1061.
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Scheme 14 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

Suppliers (76)

Suppliers (3)

31-614-CAS-36765089

Steps: 1 Yield: 100%

Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders

By: Minneci, Marco; et al

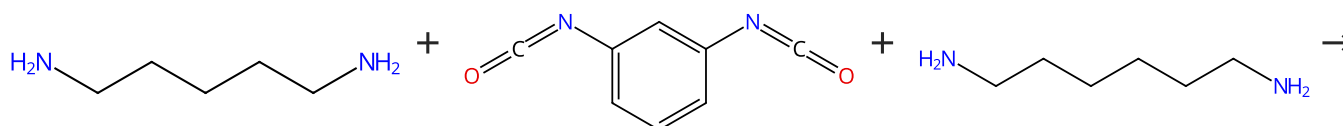
Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

1.1 Solvents: (1*S*,5*R*)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt

Experimental Protocols

Scheme 15 (1 Reaction)

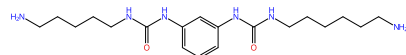
Steps: 1 Yield: 100%



Suppliers (71)

Suppliers (57)

Suppliers (81)



31-614-CAS-37221628

Steps: 1 Yield: 100%

Supramolecular protection with a recyclable molecular container: an efficient strategy for the one-pot selective functionalization of polyfunctional substrates

By: Lambert, Simon; et al

Organic Chemistry Frontiers (2023), 10(17), 4230-4242.

1.1 Reagents: Picric acid

Catalysts: 2,2',2''-[[7,15,23-Tris(1,1-dimethylethyl)-3,11,19-trioxatetracyclo[19.3.1.1^{5,9}.1^{13,17}]heptacosa-1(25),5,7,9(27),13,15,17(26),21,23-nonaene-25,26,27-triyl]tris(oxy)]tris[*N,N*-diethylacetamide]
Solvents: Chloroform-*d*, Acetonitrile-*d*₃

1.2 5 min, rt

1.3 Reagents: Picric acid

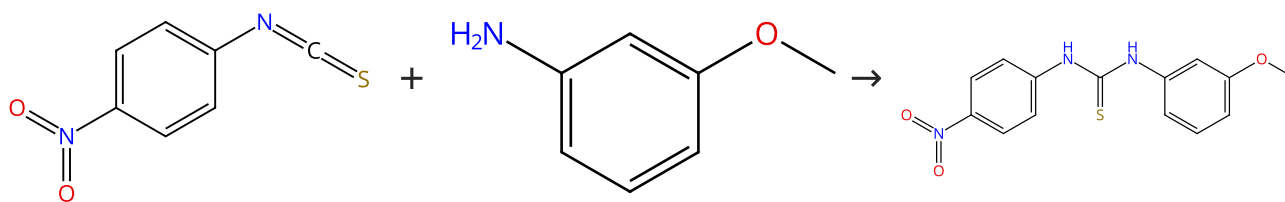
Catalysts: 2,2',2''-[[7,15,23-Tris(1,1-dimethylethyl)-3,11,19-trioxatetracyclo[19.3.1.1^{5,9}.1^{13,17}]heptacosa-1(25),5,7,9(27),13,15,17(26),21,23-nonaene-25,26,27-triyl]tris(oxy)]tris[*N,N*-diethylacetamide]
Solvents: Chloroform-*d*, Acetonitrile-*d*₃

1.4 Reagents: 1,8-Diazabicyclo[5.4.0]undec-7-ene

Experimental Protocols

Scheme 16 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

Suppliers (75)

Suppliers (2)

31-614-CAS-36765086

Steps: 1 Yield: 100%

Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders

By: Minneci, Marco; et al

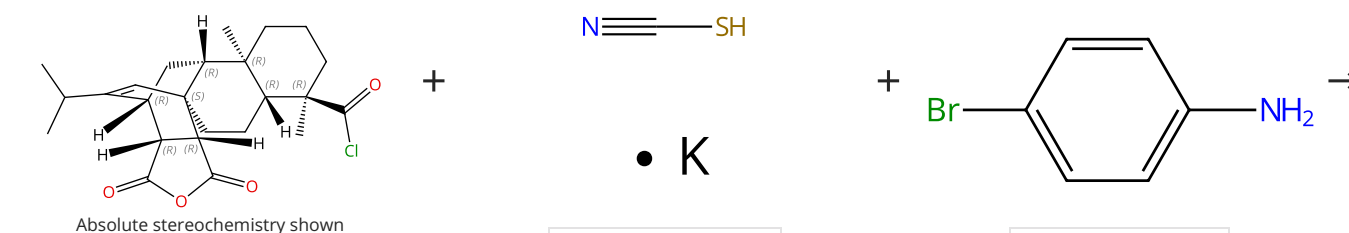
Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

1.1 Solvents: (1*S*,5*R*)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt

Experimental Protocols

Scheme 17 (1 Reaction)

Steps: 1 Yield: 100%

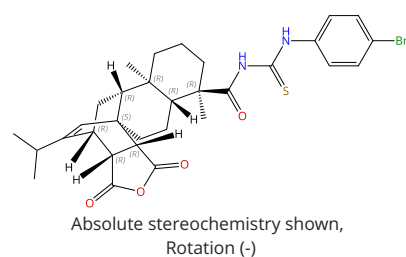


Absolute stereochemistry shown

Supplier (1)

Suppliers (116)

Suppliers (83)



31-614-CAS-34878856

Steps: 1 Yield: 100%

Synthesis of New Substituted Ureides and Thioureides Based on Maleopimaric Acid

By: Vafina, G. F.; et al

Chemistry of Natural Compounds (2022), 58(6), 1055-1061.

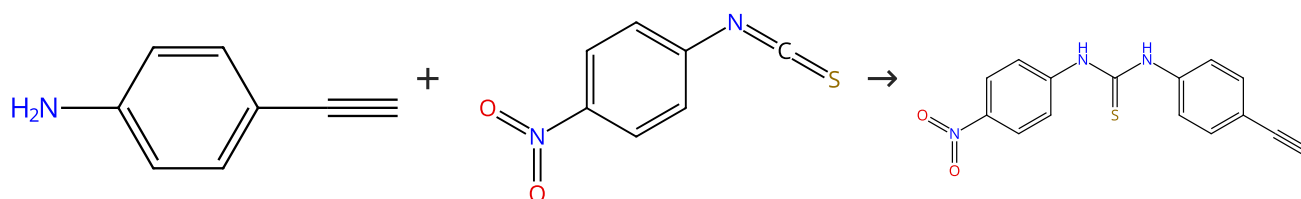
1.1 Solvents: Acetone; rt; 40 min, reflux; reflux → 5 °C

1.2 Solvents: Acetone; 0 - 5 °C; rt

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 100%



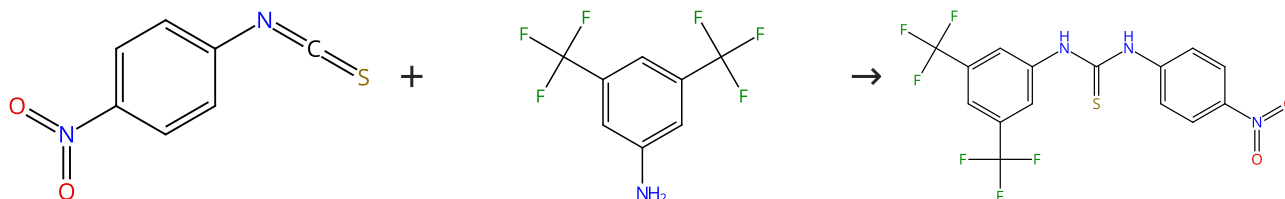
Suppliers (88)

Suppliers (75)

31-614-CAS-36765084	Steps: 1 Yield: 100%	Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders
1.1 Solvents: (1 <i>S</i> ,5 <i>R</i>)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt		By: Minneci, Marco; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

Scheme 19 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

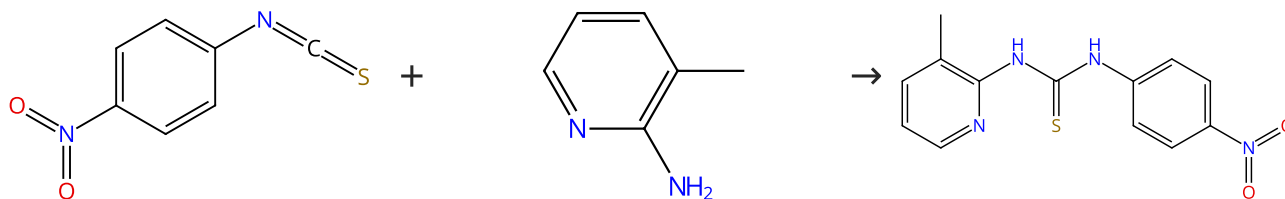
Suppliers (86)

Suppliers (2)

31-614-CAS-36765090	Steps: 1 Yield: 100%	Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders
1.1 Solvents: (1 <i>S</i> ,5 <i>R</i>)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt		By: Minneci, Marco; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

Scheme 20 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

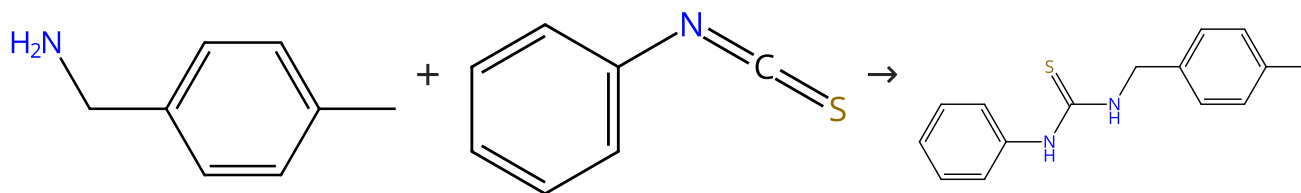
Suppliers (96)

Suppliers (3)

31-614-CAS-36765093	Steps: 1 Yield: 100%	Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders
1.1 Solvents: (1 <i>S</i> ,5 <i>R</i>)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt		By: Minneci, Marco; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

Scheme 21 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

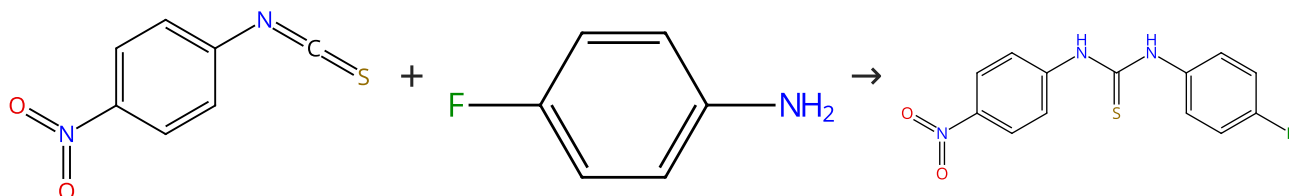
Suppliers (91)

Suppliers (31)

31-316-CAS-20805843	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas
1.1 2 - 3 min, 0 °C		By: Mandal, Subhro; et al
Experimental Protocols		Tetrahedron (2020), 76(2), 130817.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

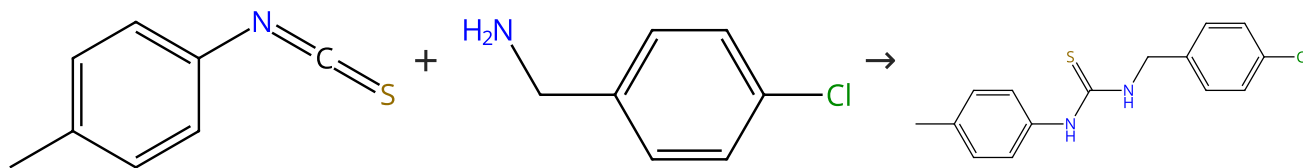
Suppliers (70)

Suppliers (2)

31-614-CAS-36765078	Steps: 1 Yield: 100%	Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders
1.1 Solvents: (1 <i>S</i> ,5 <i>R</i>)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt		By: Minneci, Marco; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

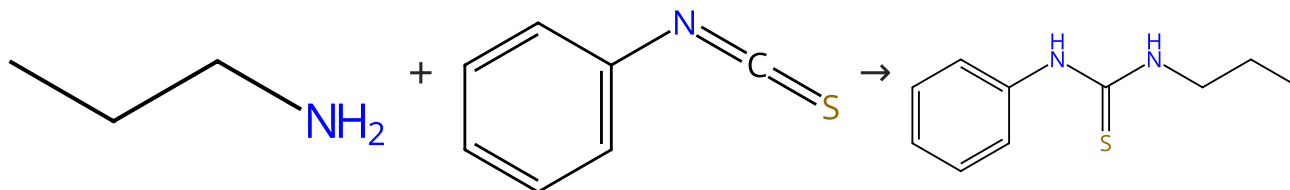
Suppliers (87)

Suppliers (4)

31-316-CAS-20805850	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas
1.1 2 - 3 min, 0 °C		By: Mandal, Subhro; et al
Experimental Protocols		Tetrahedron (2020), 76(2), 130817.

Scheme 24 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (70)

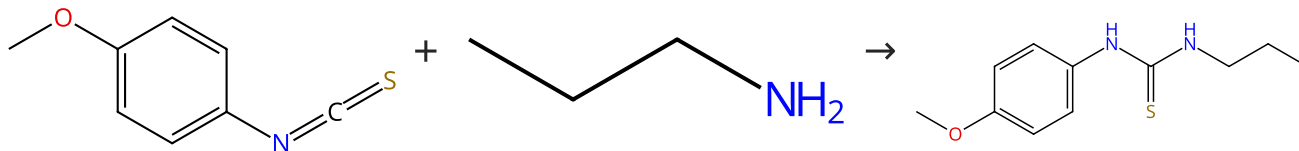
Suppliers (91)

Suppliers (41)

31-316-CAS-20805846	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 25 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (79)

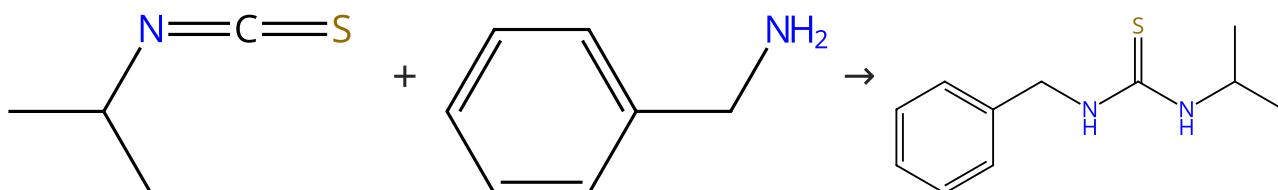
Suppliers (70)

Suppliers (7)

31-316-CAS-20805853	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 26 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (60)

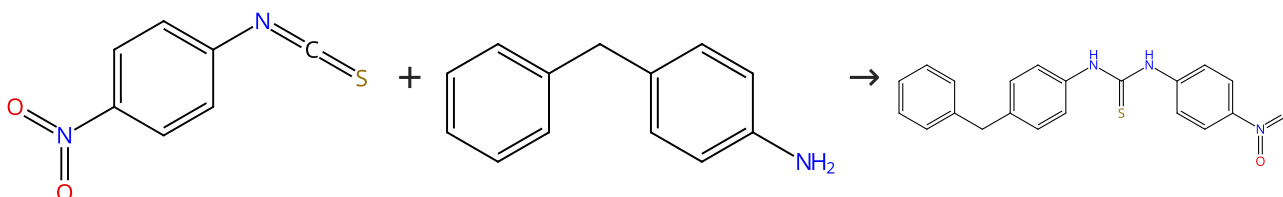
Suppliers (81)

Suppliers (5)

31-614-CAS-24399532	Steps: 1 Yield: 100%	Gram Scalable Method to Synthesize Biscarbodiimides and Asymmetric Monocarbodiimides: A Platform to Access an Array of Phosphaguanidines, Amidines, and Guanidines By: Camelio, Andrew M.; et al Journal of Organic Chemistry (2022), 87(4), 2022-2044.
1.1 Solvents: Diethyl ether; 0 °C; 12 h, 0 °C → 23 °C		
Experimental Protocols		

Scheme 27 (1 Reaction)

Steps: 1 Yield: 100%



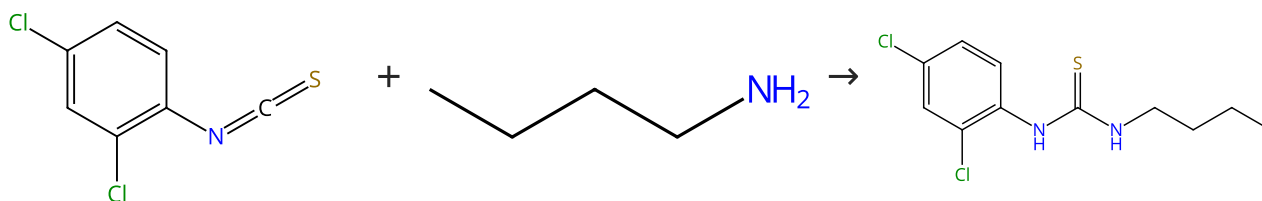
Suppliers (75)

Suppliers (76)

31-614-CAS-36765079	Steps: 1 Yield: 100%	Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders
1.1 Solvents: (1 <i>S</i> ,5 <i>R</i>)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt		By: Minneci, Marco; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

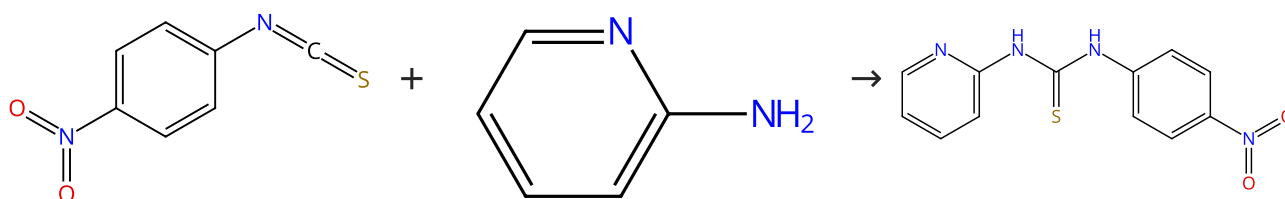
Suppliers (82)

Suppliers (3)

31-316-CAS-20805861	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas
1.1 2 - 3 min, 0 °C		By: Mandal, Subhro; et al
Experimental Protocols		Tetrahedron (2020), 76(2), 130817.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

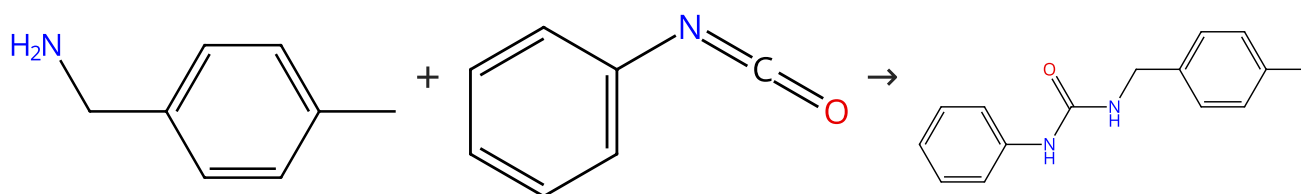
Suppliers (89)

Suppliers (4)

31-614-CAS-36765094	Steps: 1 Yield: 100%	Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders
1.1 Solvents: (1 <i>S</i> ,5 <i>R</i>)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt		By: Minneci, Marco; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (80)

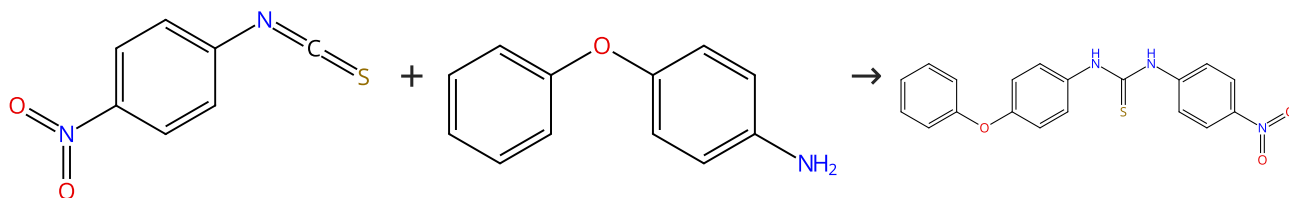
Suppliers (24)

Suppliers (6)

31-316-CAS-20805908	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas
1.1 2 - 3 min, 0 °C		By: Mandal, Subhro; et al
Experimental Protocols		Tetrahedron (2020), 76(2), 130817.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

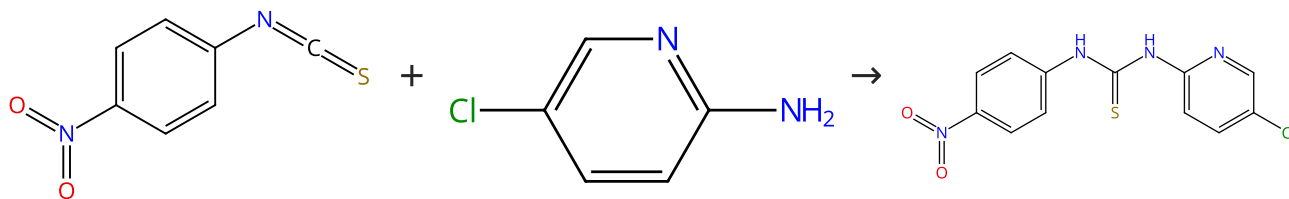
Suppliers (83)

Supplier (1)

31-614-CAS-36765076	Steps: 1 Yield: 100%	Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders
1.1 Solvents: (1 <i>S</i> ,5 <i>R</i>)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt		By: Minneci, Marco; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (75)

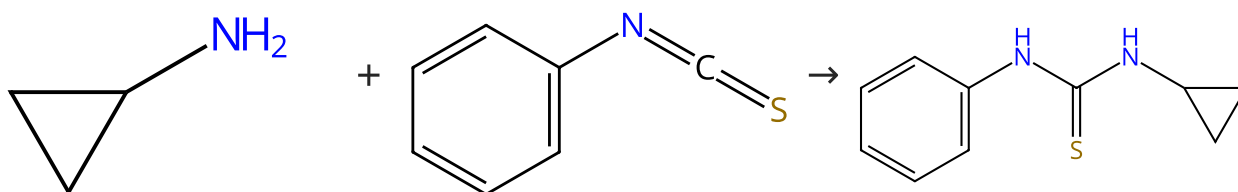
Suppliers (114)

Suppliers (4)

31-614-CAS-36765088	Steps: 1 Yield: 100%	Green synthesis of nitroaryl thioureas: Towards an improved preparation of guanidinium DNA binders
1.1 Solvents: (1 <i>S</i> ,5 <i>R</i>)-6,8-Dioxabicyclo[3.2.1]octan-4-one; 1 - 2 h, rt		By: Minneci, Marco; et al
Experimental Protocols		Bioorganic & Medicinal Chemistry Letters (2023), 90, 129346.

Scheme 33 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

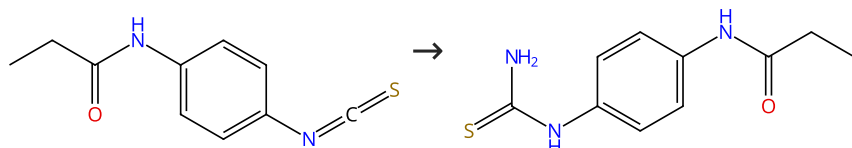
Suppliers (91)

Suppliers (14)

31-316-CAS-20805844	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 34 (1 Reaction)

Steps: 1 Yield: 100%



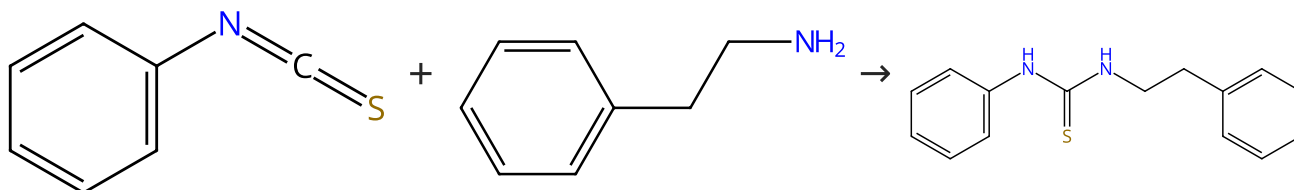
Supplier (1)

Suppliers (7)

31-316-CAS-22012499	Steps: 1 Yield: 100%	Synthesis and antibacterial evaluation of N-phenylacetamide derivatives containing 4-arylthiazole moieties By: Lu, Hui; et al Molecules (2020), 25(8), 1772.
1.1 Reagents: Ammonia Solvents: Ethanol; rt		

Scheme 35 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (91)

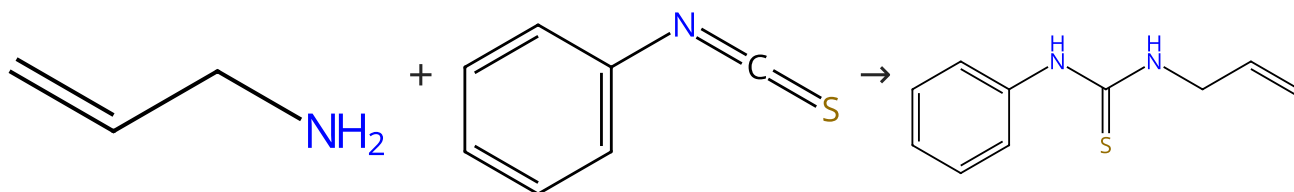
Suppliers (55)

Suppliers (52)

31-316-CAS-20805841	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 36 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (33)

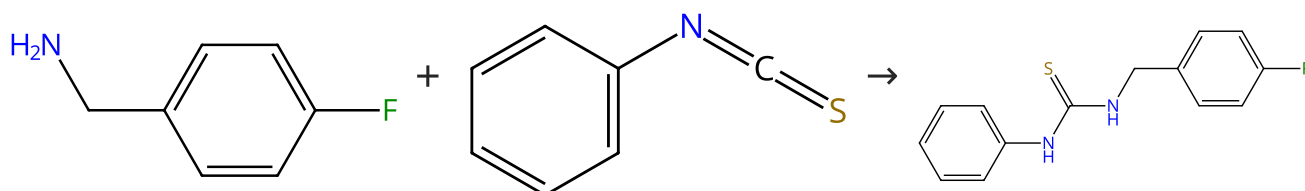
Suppliers (91)

Suppliers (33)

31-316-CAS-20805848	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 37 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (83)

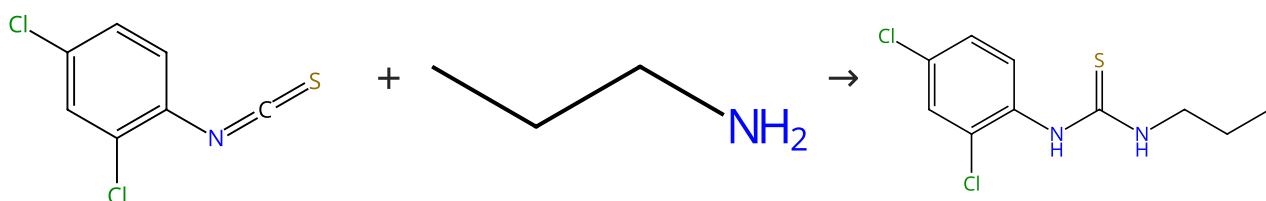
Suppliers (91)

Suppliers (15)

31-316-CAS-20805842	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 38 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

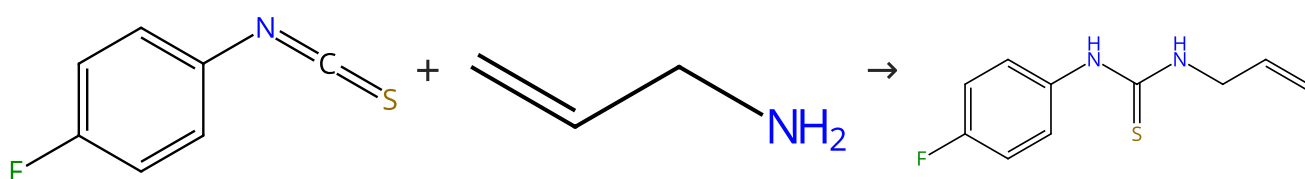
Suppliers (70)

Suppliers (5)

31-316-CAS-20805860	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 39 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (57)

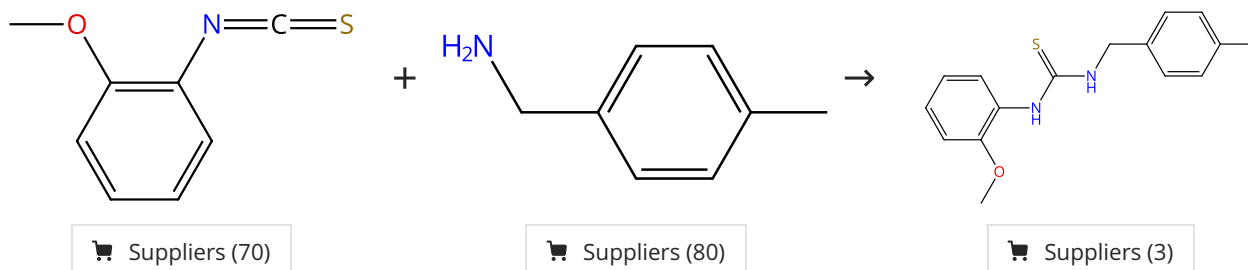
Suppliers (33)

Suppliers (8)

31-316-CAS-20805858	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 40 (1 Reaction)

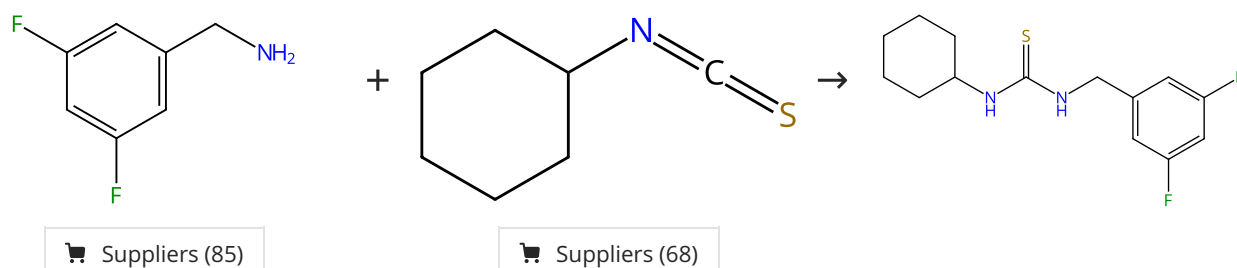
Steps: 1 Yield: 100%



31-316-CAS-20805856	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 41 (1 Reaction)

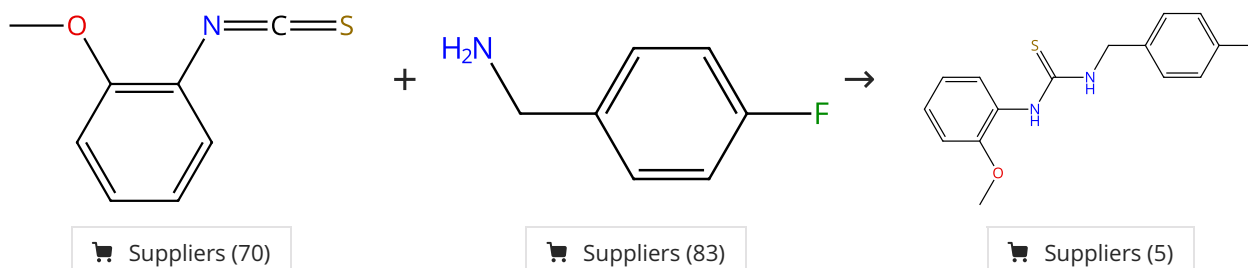
Steps: 1 Yield: 100%



31-614-CAS-24399539	Steps: 1 Yield: 100%	Gram Scalable Method to Synthesize Biscarbodiimides and Asymmetric Monocarbodiimides: A Platform to Access an Array of Phosphaguanidines, Amidines, and Guanidines By: Camelio, Andrew M.; et al Journal of Organic Chemistry (2022), 87(4), 2022-2044.
1.1 Solvents: Dichloromethane; 0 °C; 20 h, 23 °C		
Experimental Protocols		

Scheme 42 (1 Reaction)

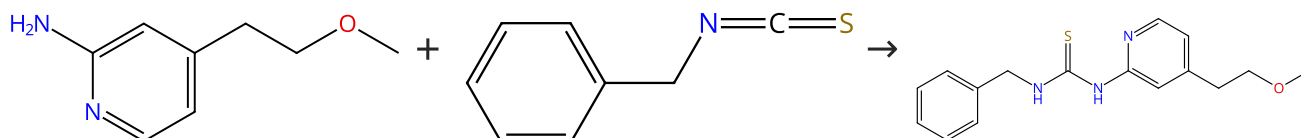
Steps: 1 Yield: 100%



31-316-CAS-20805855	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 43 (1 Reaction)

Steps: 1 Yield: 100%



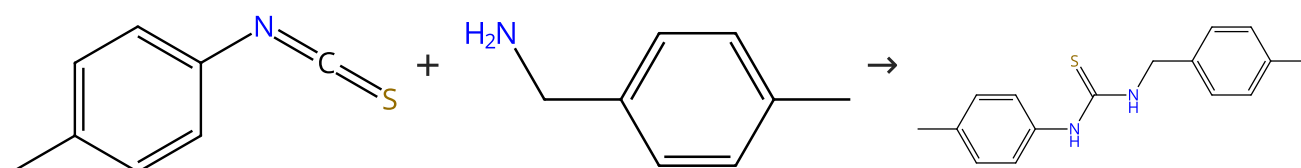
Suppliers (26)

Suppliers (91)

31-614-CAS-37134318	Steps: 1 Yield: 100%	Innovative Strategy toward Mutant CFTR Rescue in Cystic Fibrosis: Design and Synthesis of Thiadiazole Inhibitors of the E3 Ligase RNF5 By: Brusa, Irene; et al Journal of Medicinal Chemistry (2023), 66(14), 9797-9822.
1.1 Solvents: 1,2-Dichloroethane; 22 h, 55 °C		
Experimental Protocols		

Scheme 44 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (69)

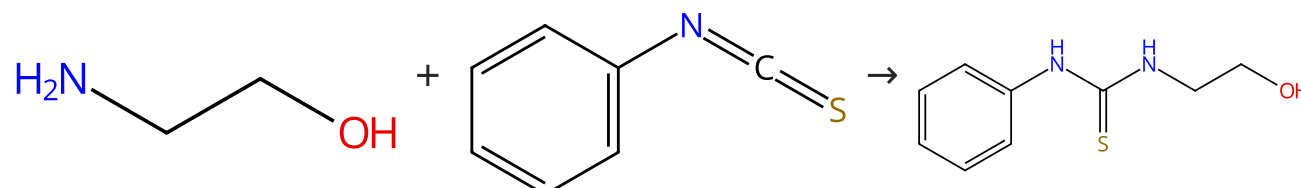
Suppliers (80)

Suppliers (2)

31-316-CAS-20805851	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 45 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (127)

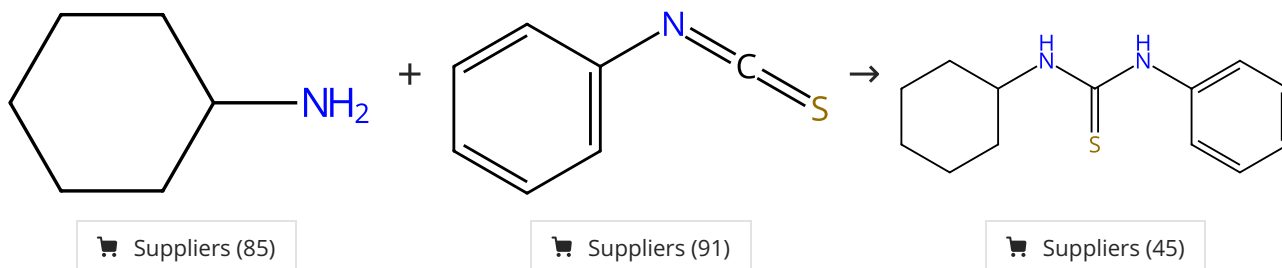
Suppliers (91)

Suppliers (53)

31-316-CAS-20805849	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 46 (1 Reaction)

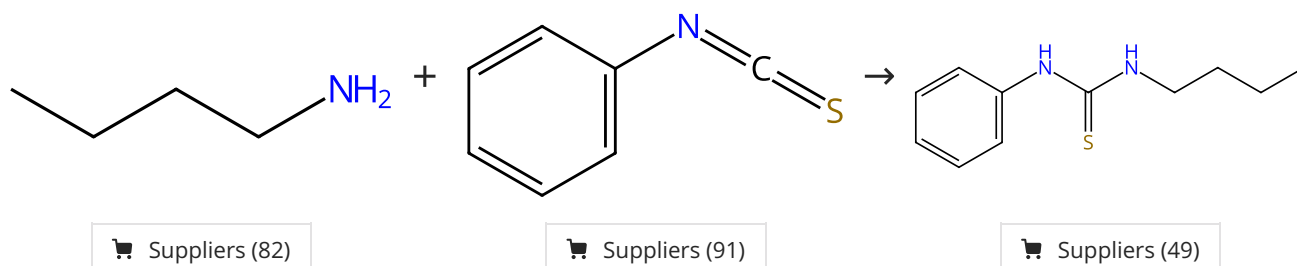
Steps: 1 Yield: 100%



31-316-CAS-20805845	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 47 (1 Reaction)

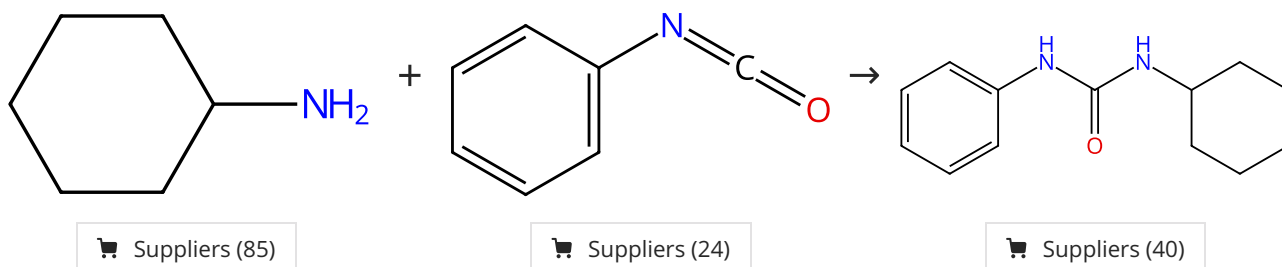
Steps: 1 Yield: 100%



31-316-CAS-20805847	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 48 (1 Reaction)

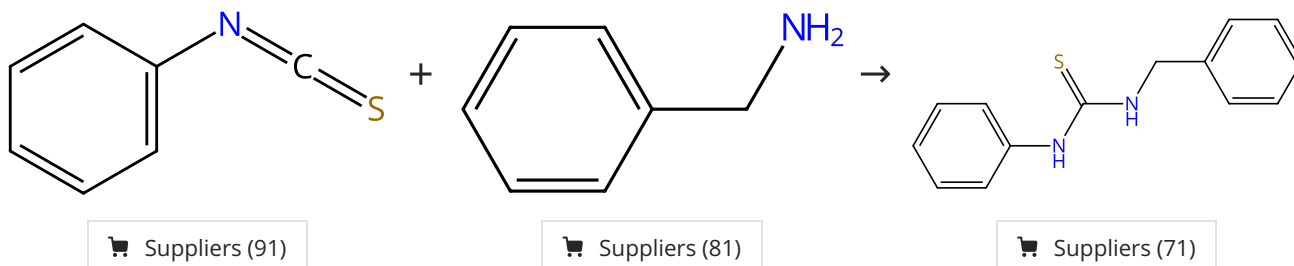
Steps: 1 Yield: 100%



31-316-CAS-20805910	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 49 (1 Reaction)

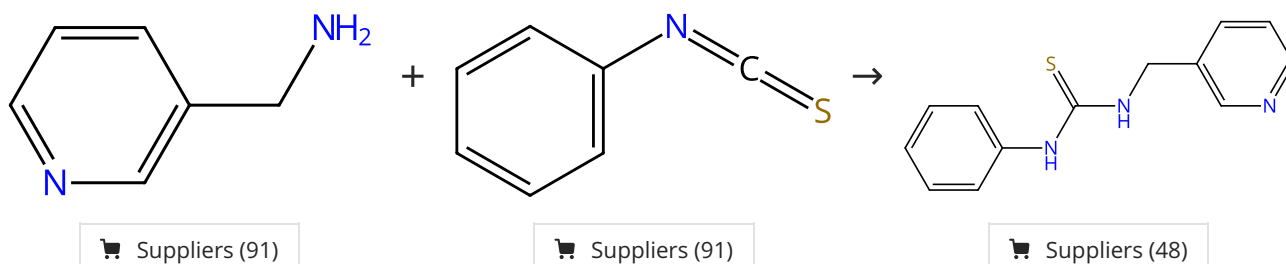
Steps: 1 Yield: 100%



31-316-CAS-20805839	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		

Scheme 50 (1 Reaction)

Steps: 1 Yield: 100%



31-316-CAS-20805840	Steps: 1 Yield: 100%	Facile synthesis of phthalidyl fused spiro thiohyd antoins through silica sulfuric acid induced oxidative rearrangement of ninhydrin adducts of thioureas By: Mandal, Subhro; et al Tetrahedron (2020), 76(2), 130817.
1.1 2 - 3 min, 0 °C		
Experimental Protocols		